

Zero inflated models

Data: College drinking

Survey data from 77 college students on a dry campus (i.e., alcohol is prohibited) in the US. Survey asks students "How many alcoholic drinks did you consume last weekend?"

- + drinks: the number of drinks the student reports consuming
- + sex: an indicator for whether the student identifies as male
- + OffCampus: an indicator for whether the student lives off campus
- + FirstYear: an indicator for whether the student is a first-year student

Our goal: model the number of drinks students report consuming.

Class activity: The fitted model

$$P(Y_i = y) = \begin{cases} e^{-\lambda_i}(1 - \alpha_i) + \alpha_i & y = 0 \\ \frac{e^{-\lambda_i} \lambda_i^y}{y!} (1 - \alpha_i) & y > 0 \end{cases}$$

$$\log\left(\frac{\hat{\alpha}_i}{1 - \hat{\alpha}_i}\right) = -0.60 + 1.14 \textit{FirstYear}_i$$

$$\log(\hat{\lambda}_i) = 0.75 + 0.42 \textit{OffCampus}_i + 1.02 \textit{Male}_i$$

What is the estimated probability that a first year student never drinks?

$$\log\left(\frac{\hat{\alpha}_i}{1 - \hat{\alpha}_i}\right) = -0.60 + 1.14$$

$$\Rightarrow \hat{\alpha}_i = 0.63$$

The fitted model

$$P(Y_i = y) = \begin{cases} e^{-\lambda_i}(1 - \alpha_i) + \alpha_i & y = 0 \\ \frac{e^{-\lambda_i} \lambda_i^y}{y!} (1 - \alpha_i) & y > 0 \end{cases}$$

$$\log\left(\frac{\hat{\alpha}_i}{1 - \hat{\alpha}_i}\right) = -0.60 + 1.14 \textit{FirstYear}_i$$

$$\log(\hat{\lambda}_i) = 0.75 + 0.42 \textit{OffCampus}_i + 1.02 \textit{Male}_i$$

What is the estimated average number of drinks for a male student who lives off campus and sometimes drinks?

$$\hat{\lambda}_i = \exp\{0.75 + 0.42 + 1.02\} = 8.93$$

The fitted model

$$\hat{\alpha}_i = 0.63$$

$$\hat{\lambda}_i = 8.93$$

$$P(Y_i = y) = \begin{cases} e^{-\lambda_i}(1 - \alpha_i) + \alpha_i & y = 0 \\ \frac{e^{-\lambda_i} \lambda_i^y}{y!} (1 - \alpha_i) & y > 0 \end{cases}$$

$$\log\left(\frac{\hat{\alpha}_i}{1 - \hat{\alpha}_i}\right) = -0.60 + 1.14 \text{FirstYear}_i$$

$$\log(\hat{\lambda}_i) = 0.75 + 0.42 \text{OffCampus}_i + 1.02 \text{Male}_i$$

What is the estimated probability that a male first year student who lives off campus had at least one drink last weekend?

$$P(Y_i > 0) = 1 - P(Y_i = 0)$$

$$P(Y_i = 0) = e^{-\hat{\lambda}_i} (1 - \hat{\alpha}_i) + \hat{\alpha}_i = 0.37$$

$$\Rightarrow P(Y_i > 0) = 1 - (e^{-8.93} (1 - 0.63) + 0.63)$$

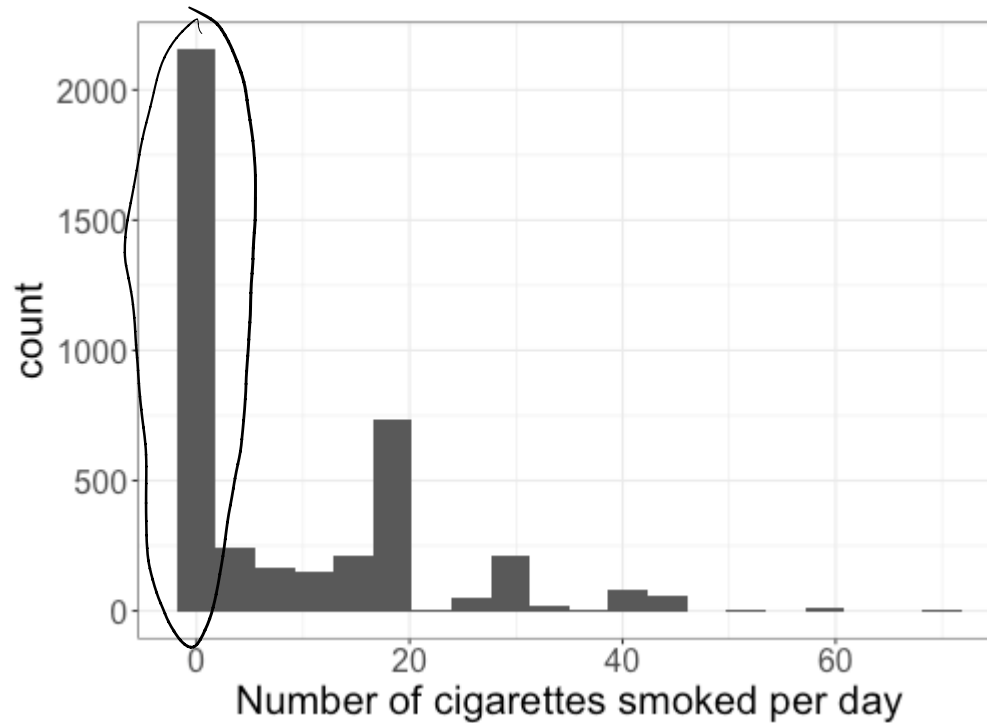
Data: Framingham heart study

Data collected on residents of Framingham, MA over a long period of time, to study variables related to heart health. We will work with a subset of the data, containing

- + `cigsPerDay`: The number of cigarettes smoked per day during the study period.
- + `education`: 1 = High School, 2 = Some College, 3 = College Degree, 4 = Advanced Degree.
- + `male`: 1 = Male, 0 = Female.
- + `age`: The age of the individual in years.
- + `diabetes`: 1 if the individual has diabetes, 0 otherwise.

Why might we see zero inflation in the number of cigarettes smoked per day?

EDA: number of cigarettes smoked



Class activity

Work on the class activity (handout).

Class activity

$$P(Y_i = y) = \begin{cases} e^{-\lambda_i}(1 - \alpha_i) + \alpha_i & y = 0 \\ \frac{e^{-\lambda_i} \lambda_i^y}{y!} (1 - \alpha_i) & y > 0 \end{cases}$$

$$\log\left(\frac{\hat{\alpha}_i}{1 - \hat{\alpha}_i}\right) = -2.51 + 0.051 \text{Age}_i$$

$$\log(\hat{\lambda}_i) = 3.21 - 0.044 \text{EducationSome}_i - 0.082 \text{EducationCollege}_i - 0.0062 \text{EducationAdv}_i - 0.024 \text{Diabetes}_i - 0.0056 \text{Age}_i$$

How do we interpret the coefficient -0.024 in the fitted model?

The expected # of cigarettes smoked for individuals who smoke w/ diabetes is about 2.4% lower than for individuals w/out diabetes who smoke, holding all other variables constant

Class activity

$$\log\left(\frac{\hat{\alpha}_i}{1 - \hat{\alpha}_i}\right) = -2.51 + 0.051Age_i$$

$$\log(\hat{\lambda}_i) = 3.21 - 0.044EducationSome_i - 0.082EducationCollege_i - 0.0062EducationAdv_i - 0.024Diabetes_i - 0.0056Age_i$$

What is the expected number of cigarettes smoked per day, for a 50 year old smoker with diabetes and some college education?

$$\begin{aligned}\hat{\lambda}_i &= \exp\{3.21 - 0.044 - 0.024 - 0.0056(50)\} \\ &\approx 17.5\end{aligned}$$

Making predictions

$$\hat{\lambda}_i = 17.5$$

$$\hat{\alpha}_i = 0.51$$

$$\log\left(\frac{\hat{\alpha}_i}{1 - \hat{\alpha}_i}\right) = -2.51 + 0.051 \text{Age}_i$$

$$\log(\hat{\lambda}_i) = 3.21 - 0.044 \text{EducationSome}_i - 0.082 \text{EducationCollege}_i - 0.0062 \text{EducationAdv}_i - 0.024 \text{Diabetes}_i - 0.0056 \text{Age}_i$$

What if I want the expected number of cigarettes smoked per day for a 50 year old with diabetes and some college education, but *without* knowing whether they are a smoker?

if $Z_i = 1$ (nonsmoker) expected # = 0

if $Z_i = 0$ (smoker) expected # = λ_i

Overall : expected # = $0 \cdot P(Z_i = 1) + \lambda_i \cdot P(Z_i = 0)$

$$= 0 \cdot \alpha_i + \lambda_i (1 - \alpha_i)$$

estimate: $\hat{\lambda}_i (1 - \hat{\alpha}_i) \Rightarrow 17.5 (1 - 0.51) = \boxed{8.6}$

A new question

$$P(Y_i = y) = \begin{cases} e^{-\lambda_i}(1 - \alpha_i) + \alpha_i & y = 0 \\ \frac{e^{-\lambda_i}\lambda_i^y}{y!}(1 - \alpha_i) & y > 0 \end{cases}$$

$$\log\left(\frac{\alpha_i}{1 - \alpha_i}\right) = \gamma_0 + \gamma_1 \text{Age}_i$$

$$\log(\lambda_i) = \beta_0 + \beta_1 \text{EducationSome}_i + \beta_2 \text{EducationCollege}_i + \beta_3 \text{EducationAdv}_i + \beta_4 \text{Diabetes}_i + \beta_5 \text{Age}_i$$

New research question: for smokers, does the number of cigarettes smoked per day depend on age?

How would we answer this research question?

$$H_0: \beta_5 = 0 \quad H_A: \beta_5 \neq 0$$

Fitting the model in R

function to fit ZIP model (Psc1 package)
response variable

```
m2 <- zeroinfl(cigsPerDay ~ education +  
               diabetes + age | age,  
               data = heart_data)
```

```
summary(m2)
```

logistic component

Count model coefficients (poisson with log link):

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	3.2063376	0.0342290	93.673	< 2e-16	***
education2	-0.0441182	0.0124809	-3.535	0.000408	***
education3	-0.0820372	0.0158604	-5.172	2.31e-07	***
education4	-0.0062453	0.0171640	-0.364	0.715964	
diabetes	-0.0241459	0.0386337	-0.625	0.531974	
age	-0.0056182	0.0006738	-8.338	< 2e-16	***

Zero-inflation model coefficients (binomial with logit link):

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-2.506681	0.191483	-13.09	<2e-16	***
age	0.051329	0.003821	13.43	<2e-16	***

z-test

```
m2 <- zeroinfl(cigsPerDay ~ education +  
               diabetes + age | age,  
               data = heart_data)  
  
summary(m2)
```

Count model coefficients (poisson with log link):

	Estimate	Std. Error	z value	Pr(> z)	
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age	-0.0056182	0.0006738	-8.338	< 2e-16	***

$$H_0: \beta_s = 0 \quad H_A: \beta_s \neq 0$$

$$z = -8.338$$

$$p\text{-value} \approx 0$$

Likelihood ratio test

Test statistic: $2 \log \left(\frac{L_{\text{full}}}{L_{\text{reduced}}} \right)$

$$= 2 (\log L_{\text{full}} - \log L_{\text{reduced}})$$

Under H_0 : $\sim \chi^2_q$ $q = \#$ parameters tested

Likelihood ratio test

```
m2 <- zeroinfl(cigsPerDay ~ education +  
               diabetes + age | age,  
               data = heart_data)  
  
m2$loglik
```

← full model

```
## [1] -14023.42
```

```
m1 <- zeroinfl(cigsPerDay ~ education +  
               diabetes | age,  
               data = heart_data)  
  
m1$loglik
```

← reduced model

```
## [1] -14058.41
```

Likelihood ratio test

```
m1 <- zeroinfl(cigsPerDay ~ education +  
               diabetes | age,  
               data = heart_data)
```

```
m2 <- zeroinfl(cigsPerDay ~ education +  
               diabetes + age | age,  
               data = heart_data)
```

```
2*(m2$loglik - m1$loglik)
```

```
## [1] 69.97593
```

```
pchisq(69.976, df=1, lower.tail=F)
```

```
## [1] 6.003042e-17
```

≈ 0

Class activity

Work on the class activity. At the end of class, render your work as an HTML and submit on Canvas.